Comparing Air Pollution Trends Between Delhi, India and Dhaka, Bangladesh from 2003-2023

ENV 872L Final Project

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Introduction

Air pollution is one of the most pressing environmental and public health challenges of the 21st century, particularly across South Asia, where rapid urbanization, industrial growth, vehicular emissions, and agricultural waste burning have made several cities among the most polluted in the world (Ditta et al., 2026). Delhi, the capital of India, and Dhaka, the capital of Bangladesh, are two such megacities that consistently

rank among the worst for air quality globally, with both cities experiencing severe seasonal pollution cycles driven by winter temperature inversions, stubble burning, brick kilns, and vehicle emissions along the shared Indo-Gangetic Plain (Ditta et al., 2026; Begum et al., 2018). Despite their geographic proximity and shared pollution sources, the two cities have followed divergent regulatory paths. India launched its National Clean Air Programme in 2019 targeting significant reductions in particulate matter (Garg et al., 2022), while Bangladesh was among the first South Asian nations to ban two-stroke engine vehicles in the early 2000s and transition to CNG-powered alternatives (State of Global Air, n.d.), raising important questions about whether these differing policy trajectories have translated into meaningfully different pollution trends over time. While prior studies have individually examined air quality trends in Delhi (Garg et al., 2022; Singh et al., 2023) and Dhaka (Islam et al., 2021; Begum & Hopke, 2018), long-term comparative analyses spanning both cities remain limited. This project addresses that gap by using 20 years of monthly air quality data (2003 to 2023) to compare pollution trends in PM2.5, PM10, ozone (O3), and AQI across both cities, testing whether air quality exhibits a significant seasonal component or decreasing trend, whether Delhi's pollution is significantly worse than Dhaka's, and whether the two cities are improving at statistically different rates.

Data description

Methods

Results

Discussion

Limitations

Conclusion

References

What to focus on/ DELETE FOR FINAL OUTPUT

```
# Limiting data from January 1st, 2003 to December 31st, 2023 because we don't have full year's data fo
# Filter for columns (6): Location, Date/Time (in ymd format), O3, PM 2.5, PM 10, AQI
# Column names: Location, Date, O3, PM2.5, PM10, AQI
# Might need extra columns for Month and Year separately BUT AFTER INITIAL WRANGLING
```

INITIAL DATA WRANGLING

```
library(dplyr); library(lubridate); library(tidyverse);
library(here); library(zoo); library(trend);
library(Kendall); library(tseries); library(ggplot2)

here()
```

```
## [1] "/home/guest/872L/ShuklaRahman"
```

```r
Delhi_dataset_raw <- read.csv(here(
  "DelhiDataset/delhi_air_quality_feature_store_processed.csv"),
  stringsAsFactors = TRUE)

Delhi_dataset_summarized <- Delhi_dataset_raw %>%
  select(city, event_timestamp, pm25, pm10, o3, aqi) %>%
  mutate(event_timestamp = ymd_hms(event_timestamp)) %>%
  filter(event_timestamp >= ymd("2003-01-01") &
    event_timestamp <= ymd("2023-12-31")) %>%
  mutate(month = month(event_timestamp),
         year = year(event_timestamp)) %>%
  group_by(month, year) %>%
  summarize(
    PM2.5 = mean(pm25, na.rm = TRUE),
    PM10 = mean(pm10, na.rm = TRUE),
    O3 = mean(o3, na.rm = TRUE),
    AQI = mean(aqi, na.rm = TRUE)
  ) %>%
  mutate(Location = "Delhi")

write.csv(Delhi_dataset_summarized,
          here("processed/delhi_final_processed.csv"), row.names = FALSE)
```

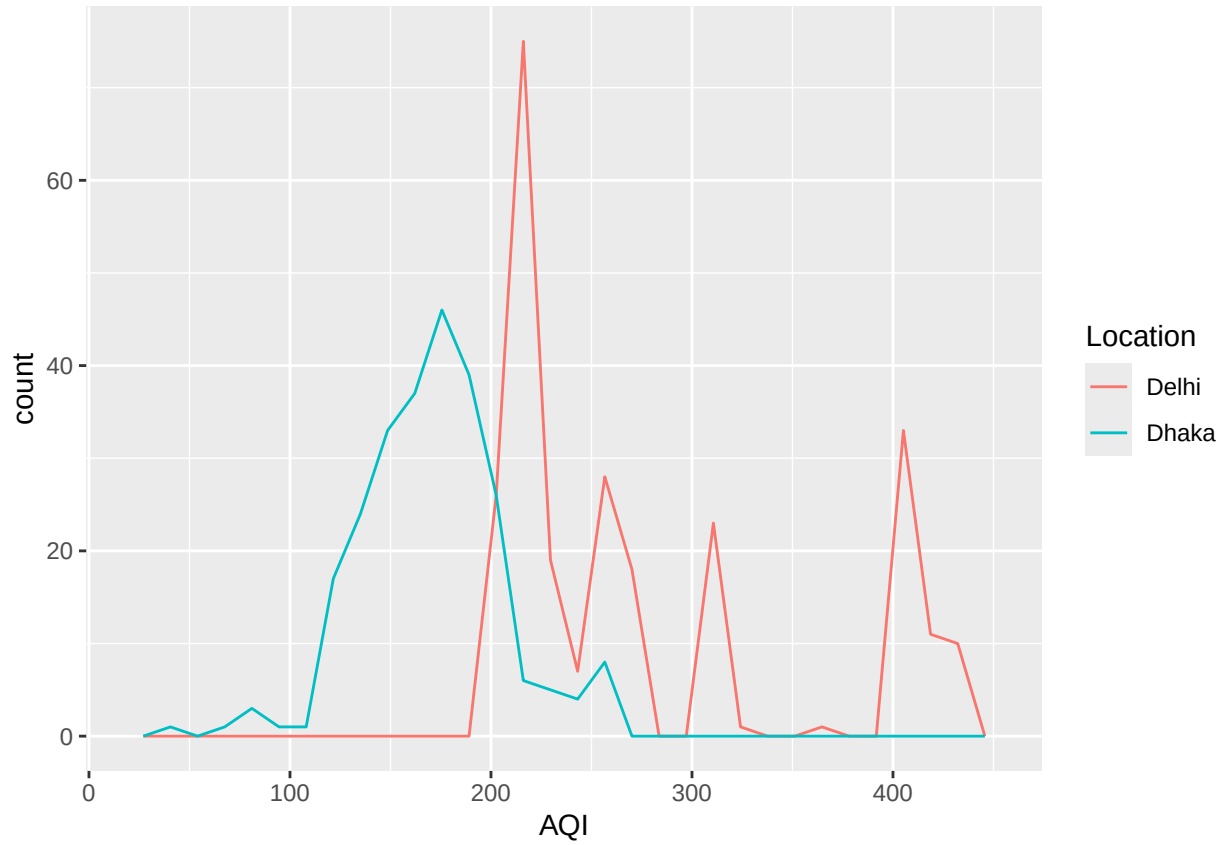
```r
#reading csv file
BD_dataset_raw <- read.csv(here("AQI Bangladesh.csv"),
        stringsAsFactors = TRUE)

#filtering for Dhaka only
dhaka_dataset_raw <- BD_dataset_raw %>%
  filter(city_name == "Dhaka")

view(dhaka_dataset_raw)

#filtered for date,location,pm2.5,pm10,aqi,o3
dhaka_air <- dhaka_dataset_raw %>%
  mutate(datetime = ymd_hm(datetime)) %>%
  filter(city_name == "Dhaka") %>%
  select(city_name, datetime, ozone, pm2_5, pm10, aqi) %>%
  rename(
    Location  = city_name,
    Date = datetime,
    O3        = ozone,
    PM2.5     = pm2_5,
    PM10      = pm10,
    AQI       = aqi
  )

# Filter date range
dhaka_air <- dhaka_air %>%
  filter(Date >= as.Date("2003-01-01") & Date <= as.Date("2023-12-31"))

dhaka_dataset_summarized <- dhaka_air %>%
  mutate(month = month(Date),
```

```r
         year  = year(Date)) %>%
  group_by(month, year) %>%
  summarize(
    PM2.5 = mean(PM2.5, na.rm = TRUE),
    PM10  = mean(PM10,  na.rm = TRUE),
    O3    = mean(O3,    na.rm = TRUE),
    AQI   = mean(AQI,   na.rm = TRUE)
  ) %>%
  mutate(Location = "Dhaka")

view(dhaka_dataset_summarized)

write.csv(dhaka_dataset_summarized,
          here("processed/dhaka_final_processed.csv"), row.names = FALSE)
```

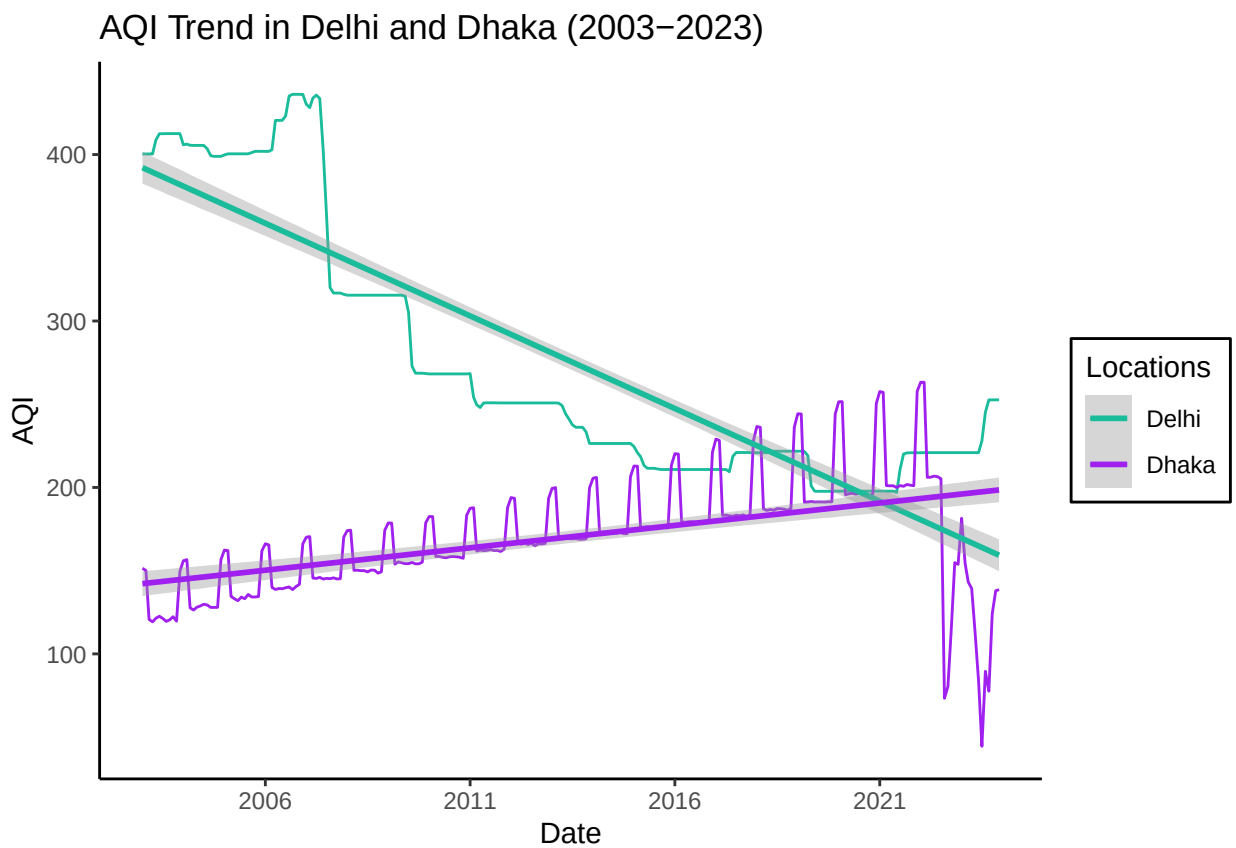
```r
#reading both datasets
dhaka_final_processed <- read.csv(here("processed/dhaka_final_processed.csv"))
delhi_final_processed <- read.csv(here("processed/delhi_final_processed.csv"))

#combining into one dataframe
combined_dataset <- rbind(dhaka_final_processed, delhi_final_processed)

combined_dataset <- mutate(combined_dataset,
                           first.month = make_date(year, month, 1))

#checking for missing data
#colSums(is.na(combined_dataset))
# found no NA data
```

TIME SERIES ANALYSES

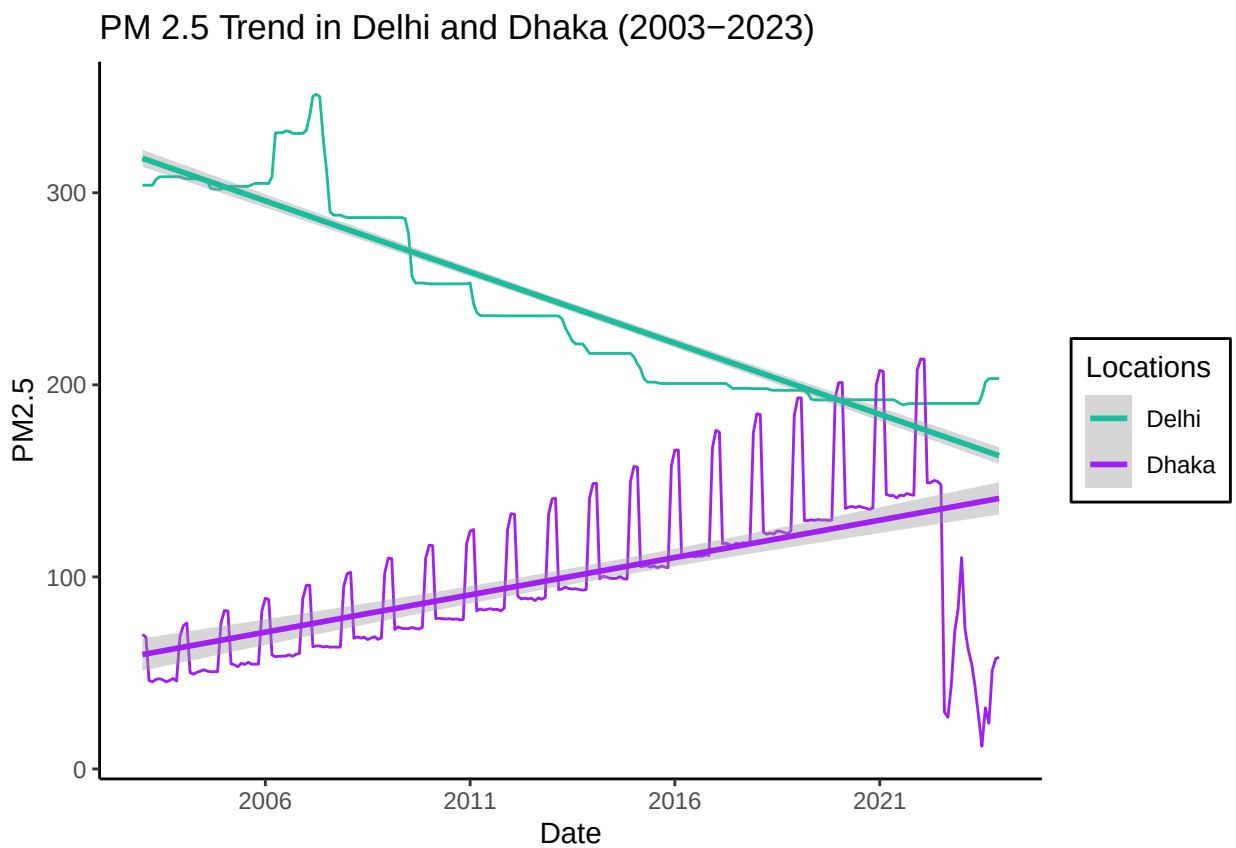
```r
# Trial Frequency Plot
ggplot(combined_dataset, aes(x = AQI, color = Location)) +
  geom_freqpoly()
```



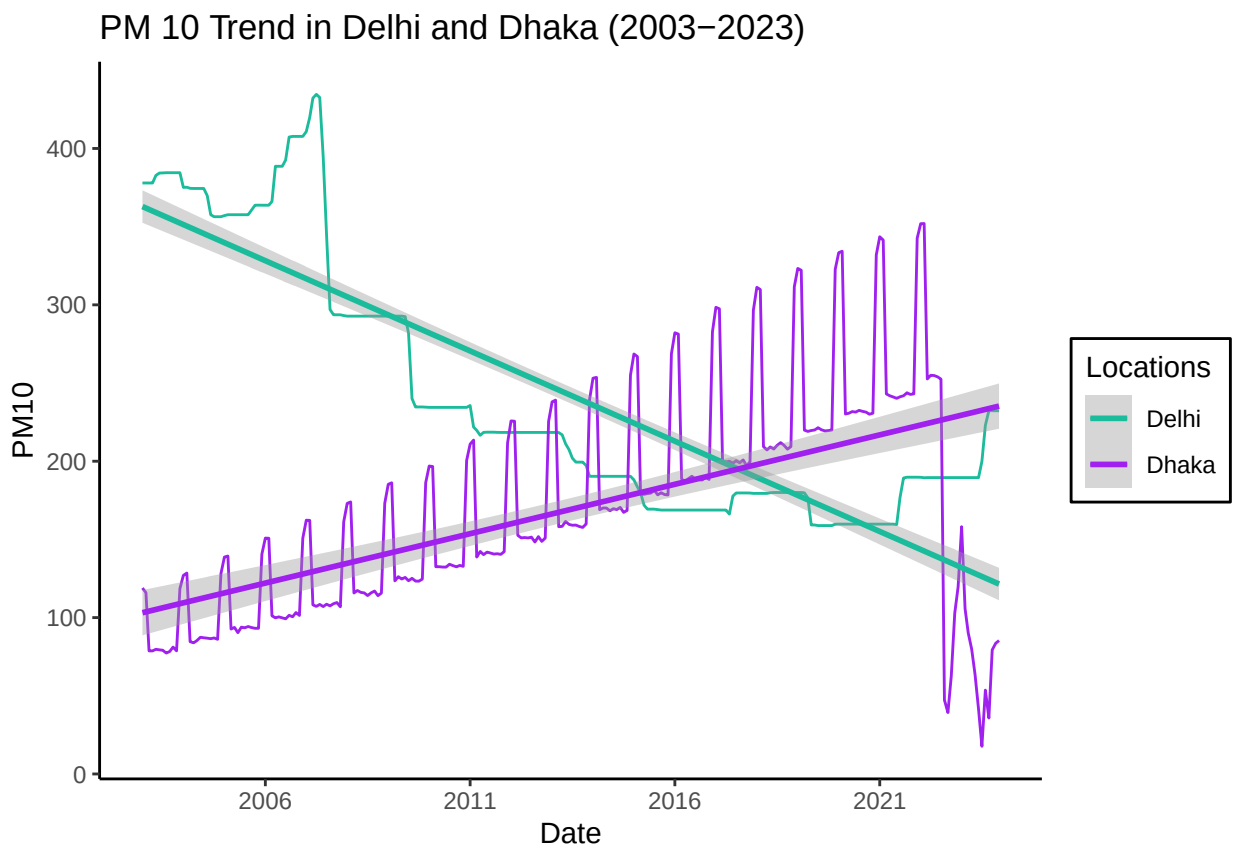

```r
#AQI
ggplot(combined_dataset, aes(y = AQI, x = first.month, colour = Location)) +
  geom_line() +
  theme_classic() +
  scale_color_manual(values = c("Delhi" = "#1abc9c", "Dhaka" = "purple")) +
  theme(legend.background = element_rect(colour = "black", fill = "white",
    linewidth = 0.5)) +
  scale_x_date(date_breaks = "5 years", date_labels = "%Y") +
  geom_smooth(method = lm) +
  labs(x = "Date", title = "AQI Trend in Delhi and Dhaka (2003-2023)",
       colour = "Locations")
```



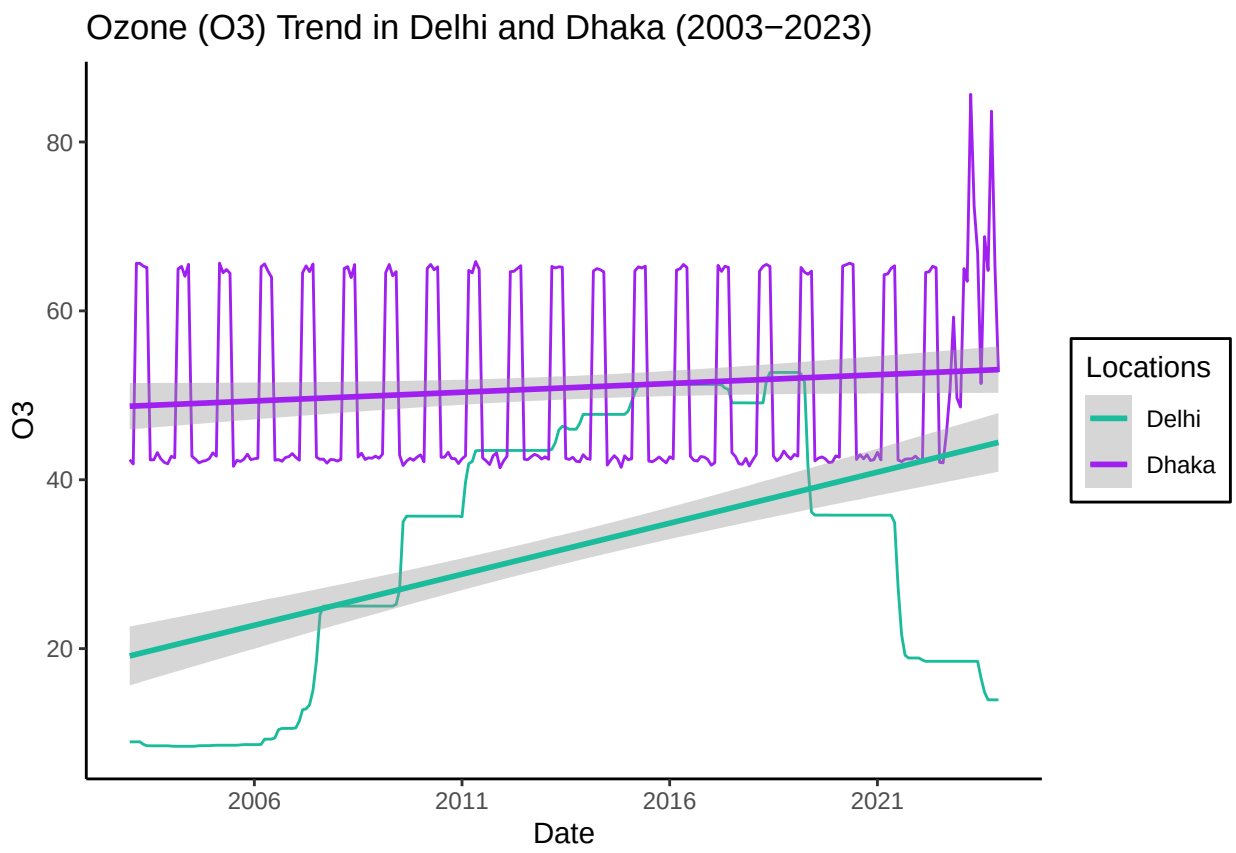

```r
#PM2.5
ggplot(combined_dataset, aes(y = PM2.5, x = first.month, colour = Location)) +
  geom_line() +
    theme_classic() +
    scale_color_manual(values = c("Delhi" = "#1abc9c", "Dhaka" = "purple")) +
    theme(legend.background = element_rect(colour = "black", fill = "white",
    linewidth = 0.5)) +
  scale_x_date(date_breaks = "5 years", date_labels = "%Y") +
  geom_smooth(method = lm) +
  labs(
    x = "Date", title = "PM 2.5 Trend in Delhi and Dhaka (2003-2023)",
    colour = "Locations")
```



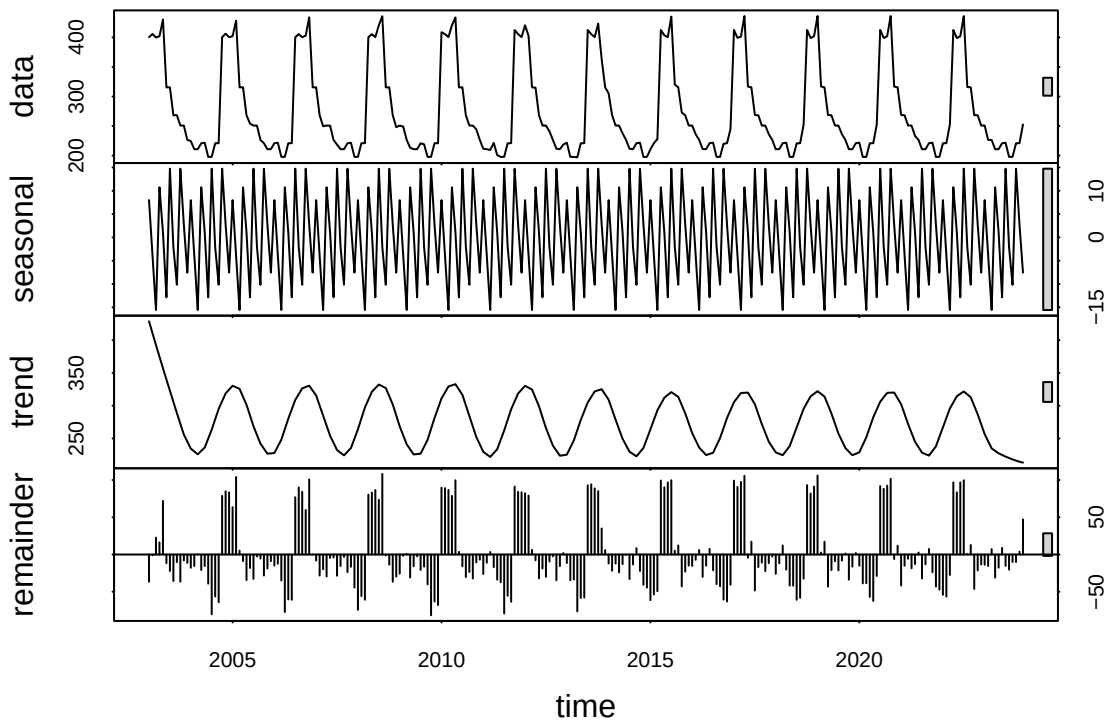

```r
#PM10
ggplot(combined_dataset, aes(y = PM10, x = first.month, colour = Location)) +
  geom_line() +
    theme_classic() +
    scale_color_manual(values = c("Delhi" = "#1abc9c", "Dhaka" = "purple")) +
    theme(legend.background = element_rect(colour = "black", fill = "white",
    linewidth = 0.5)) +
  scale_x_date(date_breaks = "5 years", date_labels = "%Y") +
  geom_smooth(method = lm) +
  labs(
    x = "Date", title = "PM 10 Trend in Delhi and Dhaka (2003-2023)",
    colour = "Locations")
```



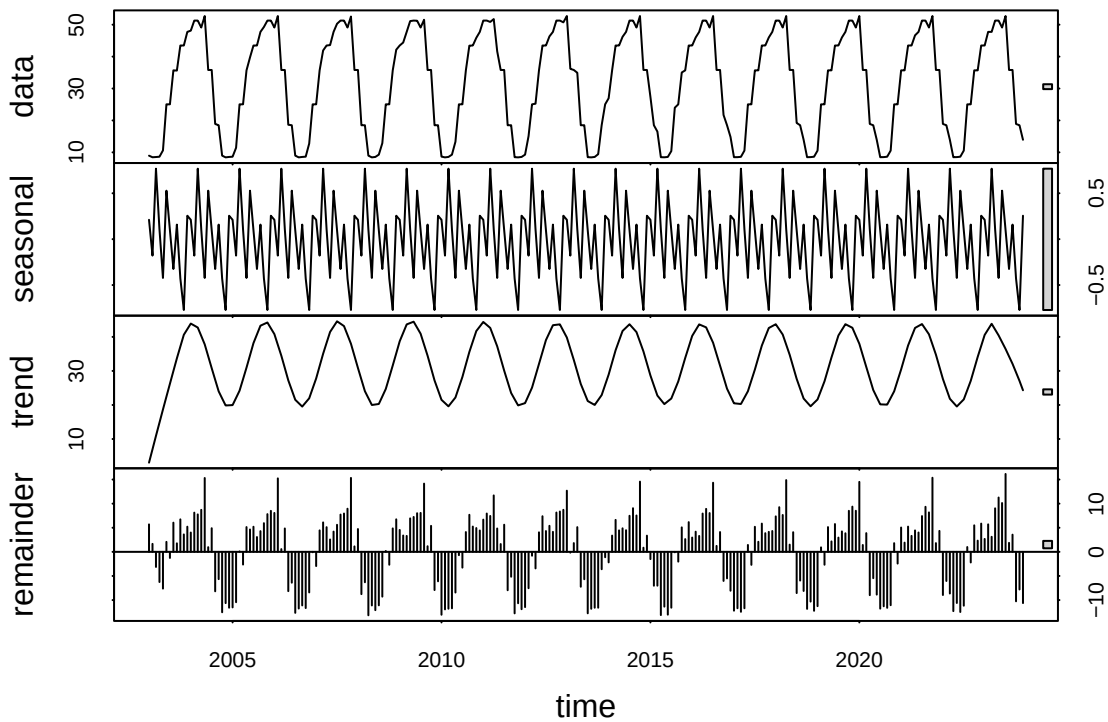

```r
#O3
ggplot(combined_dataset, aes(y = O3, x = first.month, colour = Location)) +
  geom_line() +
    theme_classic() +
    scale_color_manual(values = c("Delhi" = "#1abc9c", "Dhaka" = "purple")) +
    theme(legend.background = element_rect(colour = "black", fill = "white",
    linewidth = 0.5)) +
  scale_x_date(date_breaks = "5 years", date_labels = "%Y") +
  geom_smooth(method = lm) +
  labs(
    x = "Date", title = "Ozone (O3) Trend in Delhi and Dhaka (2003-2023)",
    colour = "Locations")
```



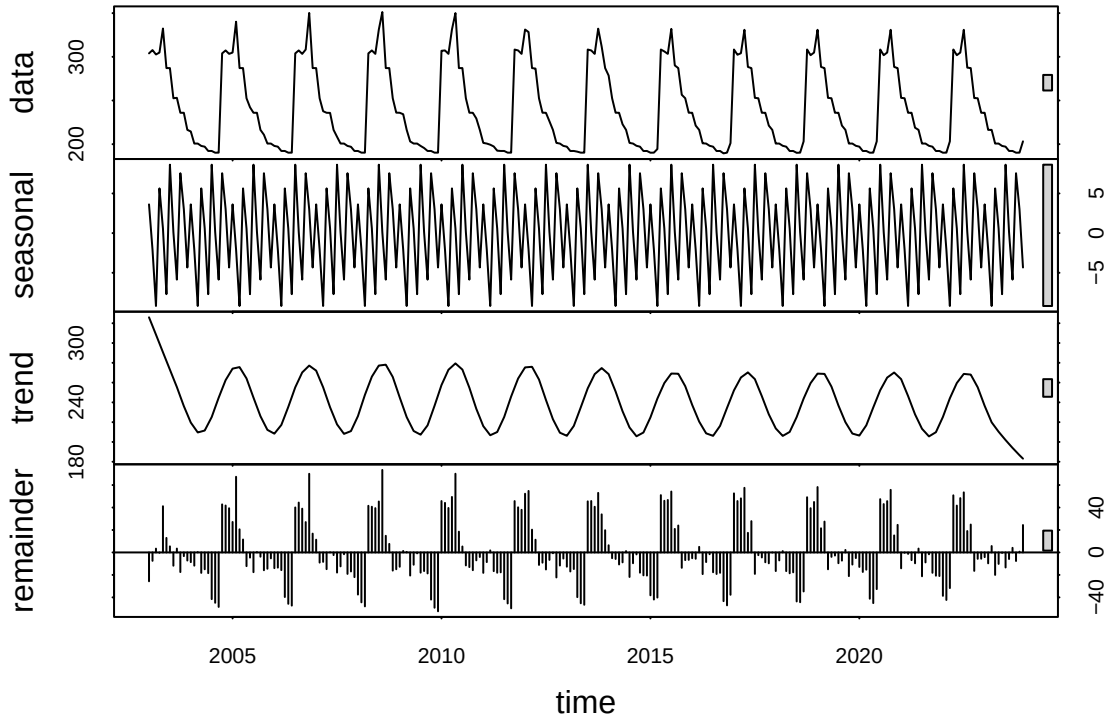

```r
# Monthly AQI for Delhi
Delhi.AQI.monthly.ts <- ts(delhi_final_processed$AQI,
                           start = c(2003,1), frequency = 12)
Delhi.AQI.monthly.decomposed <- stl(Delhi.AQI.monthly.ts,
                                    s.window = "periodic")
plot(Delhi.AQI.monthly.decomposed)
```



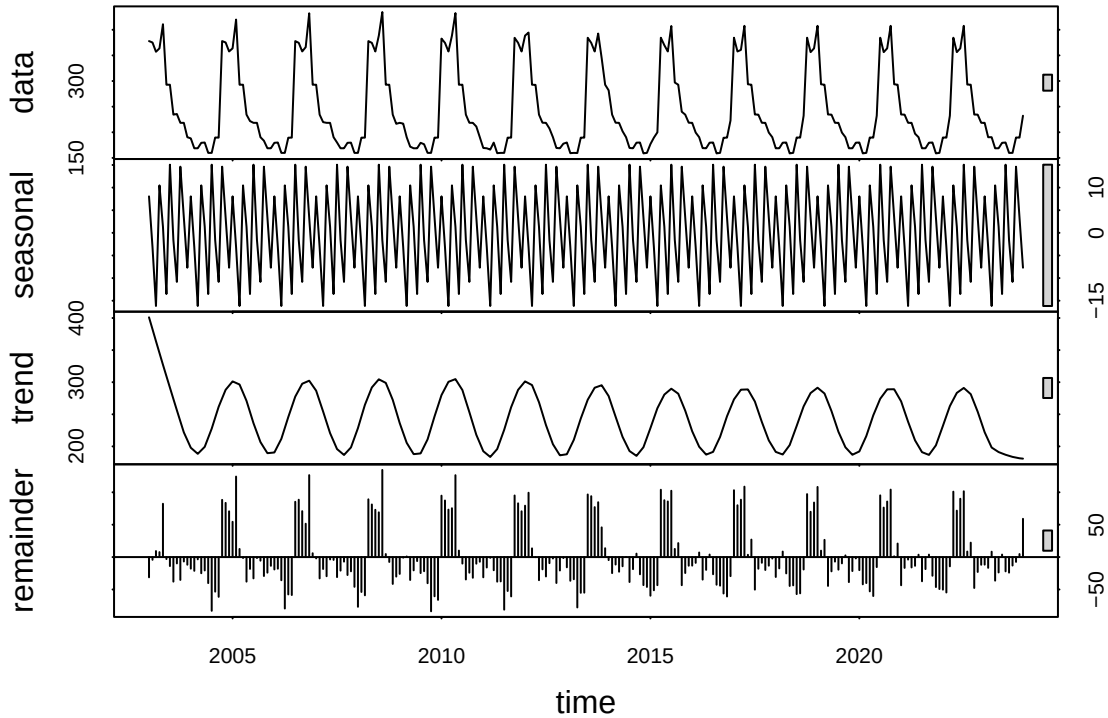

```r
# Monthly Ozone for Delhi
Delhi.Ozone.monthly.ts <- ts(delhi_final_processed$O3,
                             start = c(2003,1), frequency = 12)
Delhi.Ozone.monthly.decomposed <- stl(Delhi.Ozone.monthly.ts,
                                      s.window = "periodic")
plot(Delhi.Ozone.monthly.decomposed)
```



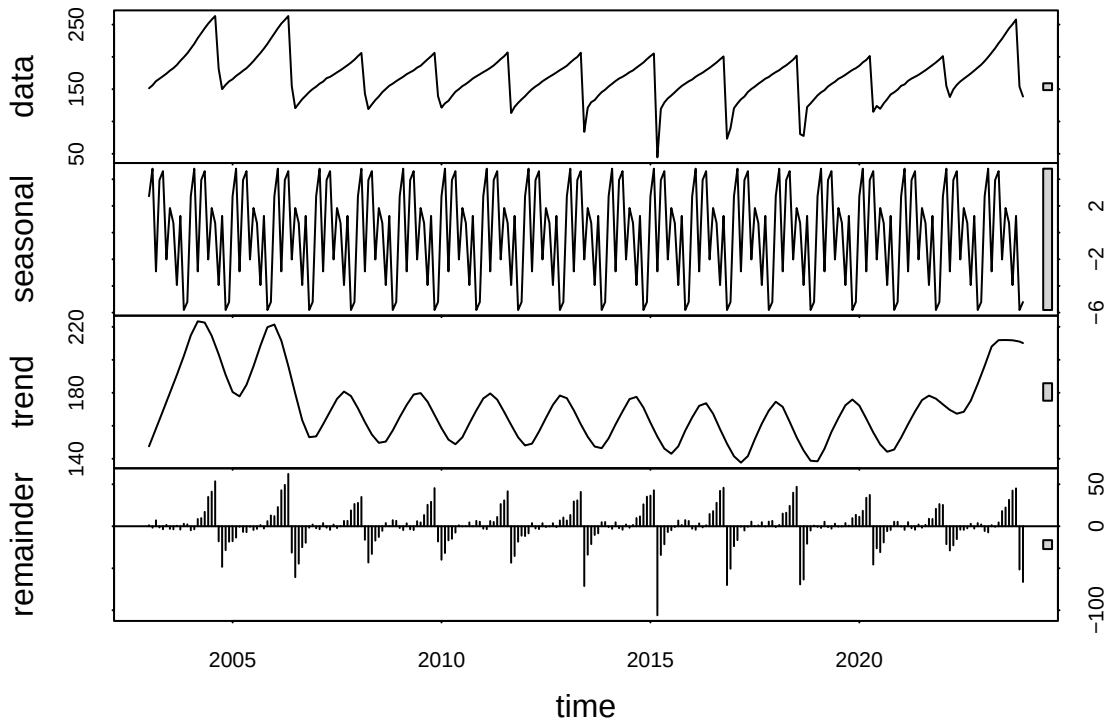

```r
# Monthly PM2.5 for Delhi
Delhi.PM2.5.monthly.ts <- ts(delhi_final_processed$PM2.5,
                             start = c(2003,1), frequency = 12)
Delhi.PM2.5.monthly.decomposed <- stl(Delhi.PM2.5.monthly.ts,
                                      s.window = "periodic")
plot(Delhi.PM2.5.monthly.decomposed)
```



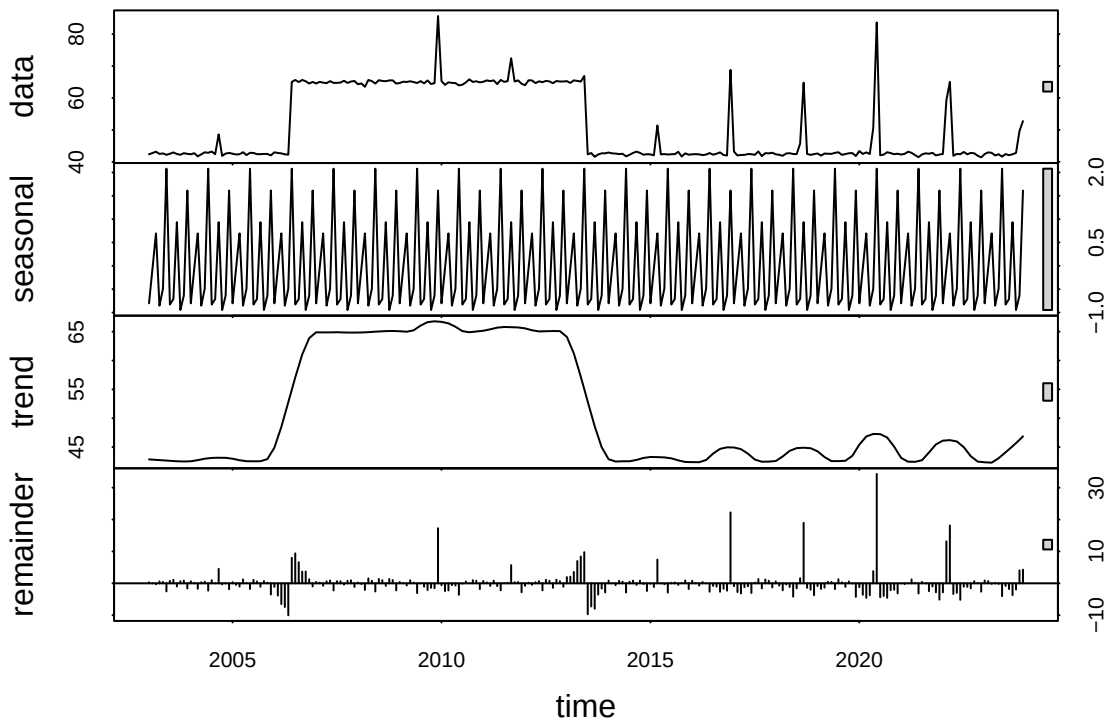

```r
# Monthly PM10 for Delhi
Delhi.PM10.monthly.ts <- ts(delhi_final_processed$PM10,
                            start = c(2003,1), frequency = 12)
Delhi.PM10.monthly.decomposed <- stl(Delhi.PM10.monthly.ts,
                                     s.window = "periodic")
plot(Delhi.PM10.monthly.decomposed)
```



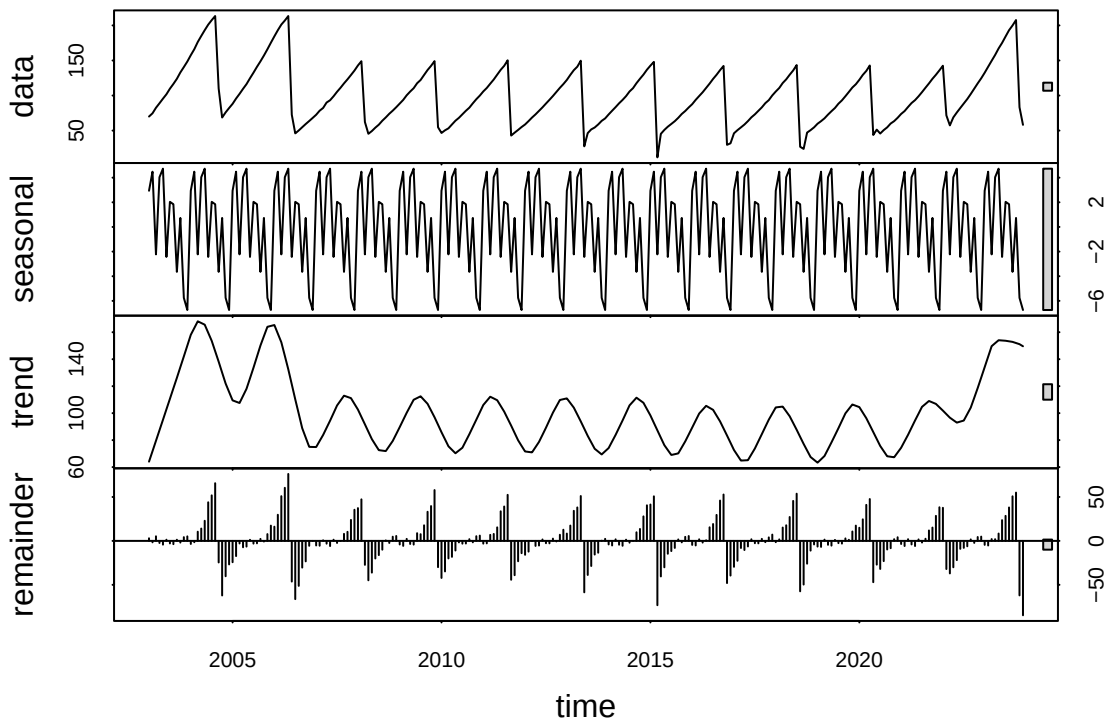

```r
# Monthly AQI for Dhaka
Dhaka.AQI.monthly.ts <- ts(dhaka_final_processed$AQI,
                           start = c(2003,1), frequency = 12)
Dhaka.AQI.monthly.decomposed <- stl(Dhaka.AQI.monthly.ts,
                                    s.window = "periodic")
plot(Dhaka.AQI.monthly.decomposed)
```



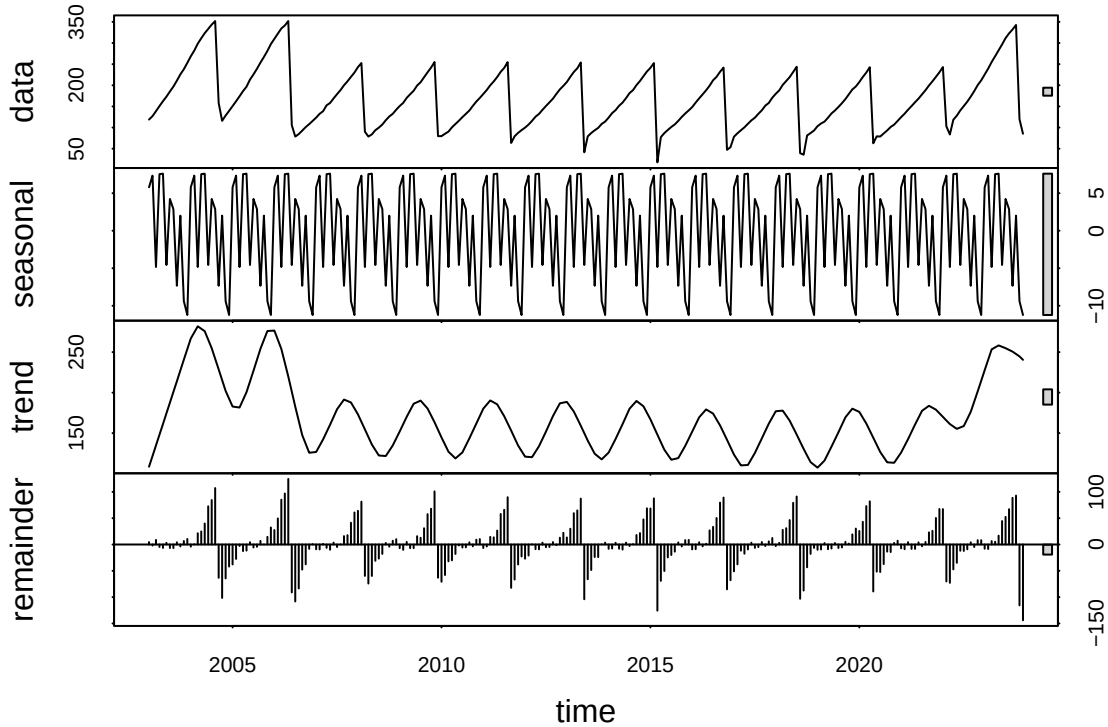

```r
# Monthly Ozone for Dhaka
Dhaka.Ozone.monthly.ts <- ts(dhaka_final_processed$O3,
                             start = c(2003,1), frequency = 12)
Dhaka.Ozone.monthly.decomposed <- stl(Dhaka.Ozone.monthly.ts,
                                      s.window = "periodic")
plot(Dhaka.Ozone.monthly.decomposed)
```




```r
# Monthly PM2.5 for Dhaka
Dhaka.PM2.5.monthly.ts <- ts(dhaka_final_processed$PM2.5,
                             start = c(2003,1), frequency = 12)
Dhaka.PM2.5.monthly.decomposed <- stl(Dhaka.PM2.5.monthly.ts,
                                      s.window = "periodic")
plot(Dhaka.PM2.5.monthly.decomposed)
```




```r
# Monthly PM10 for Dhaka
Dhaka.PM10.monthly.ts <- ts(dhaka_final_processed$PM10,
                            start = c(2003,1), frequency = 12)
Dhaka.PM10.monthly.decomposed <- stl(Dhaka.PM10.monthly.ts,
                                     s.window = "periodic")
plot(Dhaka.PM10.monthly.decomposed)
```




```r
# Seasonal Mann-Kendall Dhaka

# Dhaka AQI
Dhaka.AQI.monthly.trend <- trend::smk.test(Dhaka.AQI.monthly.ts)
Dhaka.AQI.monthly.trend
```

```
##
##  Seasonal Mann-Kendall trend test (Hirsch-Slack test)
##
## data:  Dhaka.AQI.monthly.ts
## z = -1.7696, p-value = 0.0768
## alternative hypothesis: true S is not equal to 0
## sample estimates:
##     S  varS
##  -204 13160
```

```r
summary(Dhaka.AQI.monthly.trend)
```

```
##
##  Seasonal Mann-Kendall trend test (Hirsch-Slack test)
##
## data: Dhaka.AQI.monthly.ts
## alternative hypothesis: two.sided
##
## Statistics for individual seasons
```

```
##
## H0
##                      S   varS    tau      z Pr(>|z|)
## Season 1:   S = 0   -2 1096.7 -0.010 -0.030  0.97591
## Season 2:   S = 0  -26 1096.7 -0.124 -0.755  0.45030
## Season 3:   S = 0  -14 1096.7 -0.067 -0.393  0.69464
## Season 4:   S = 0  -10 1096.7 -0.048 -0.272  0.78580
## Season 5:   S = 0  -34 1096.7 -0.162 -0.996  0.31901
## Season 6:   S = 0  -20 1096.7 -0.095 -0.574  0.56614
## Season 7:   S = 0   -6 1096.7 -0.029 -0.151  0.87999
## Season 8:   S = 0  -26 1096.7 -0.124 -0.755  0.45030
## Season 9:   S = 0   -8 1096.7 -0.038 -0.211  0.83259
## Season 10:   S = 0   6 1096.7  0.029  0.151  0.87999
## Season 11:   S = 0 -34 1096.7 -0.162 -0.996  0.31901
## Season 12:   S = 0 -30 1096.7 -0.143 -0.876  0.38119
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```r
# Dhaka AQI not significant (p = 0.0768)


# Dhaka PM2.5
Dhaka.PM2.5.monthly.trend <- trend::smk.test(Dhaka.PM2.5.monthly.ts)
Dhaka.PM2.5.monthly.trend
```

```
##
##  Seasonal Mann-Kendall trend test (Hirsch-Slack test)
##
## data:  Dhaka.PM2.5.monthly.ts
## z = -1.7696, p-value = 0.0768
## alternative hypothesis: true S is not equal to 0
## sample estimates:
##     S  varS
##  -204 13160
```

```r
summary(Dhaka.PM2.5.monthly.trend)
```

```
##
##  Seasonal Mann-Kendall trend test (Hirsch-Slack test)
##
## data: Dhaka.PM2.5.monthly.ts
## alternative hypothesis: two.sided
##
## Statistics for individual seasons
##
## H0
##                      S   varS    tau      z Pr(>|z|)
## Season 1:   S = 0    0 1096.7  0.000  0.000  1.00000
## Season 2:   S = 0  -28 1096.7 -0.133 -0.815  0.41489
## Season 3:   S = 0  -14 1096.7 -0.067 -0.393  0.69464
## Season 4:   S = 0  -12 1096.7 -0.057 -0.332  0.73976
## Season 5:   S = 0  -38 1096.7 -0.181 -1.117  0.26387
## Season 6:   S = 0  -22 1096.7 -0.105 -0.634  0.52599
```

```
## Season 7:   S = 0   -8 1096.7 -0.038 -0.211  0.83259
## Season 8:   S = 0  -26 1096.7 -0.124 -0.755  0.45030
## Season 9:   S = 0  -12 1096.7 -0.057 -0.332  0.73976
## Season 10:   S = 0   8 1096.7  0.038  0.211  0.83259
## Season 11:   S = 0 -28 1096.7 -0.133 -0.815  0.41489
## Season 12:   S = 0 -24 1096.7 -0.114 -0.695  0.48735
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```r
# Dhaka PM2.5 not significant (p = 0.0768)

# Dhaka PM10
Dhaka.PM10.monthly.trend <- trend::smk.test(Dhaka.PM10.monthly.ts)
Dhaka.PM10.monthly.trend
```

```
##
##  Seasonal Mann-Kendall trend test (Hirsch-Slack test)
##
## data:  Dhaka.PM10.monthly.ts
## z = -1.6127, p-value = 0.1068
## alternative hypothesis: true S is not equal to 0
## sample estimates:
##     S  varS
##  -186 13160
```

```r
summary(Dhaka.PM10.monthly.trend)
```

```
##
##  Seasonal Mann-Kendall trend test (Hirsch-Slack test)
##
## data: Dhaka.PM10.monthly.ts
## alternative hypothesis: two.sided
##
## Statistics for individual seasons
##
## H0
##                      S   varS    tau      z Pr(>|z|)
## Season 1:   S = 0    4 1096.7  0.019  0.091  0.92782
## Season 2:   S = 0  -26 1096.7 -0.124 -0.755  0.45030
## Season 3:   S = 0  -12 1096.7 -0.057 -0.332  0.73976
## Season 4:   S = 0  -12 1096.7 -0.057 -0.332  0.73976
## Season 5:   S = 0  -32 1096.7 -0.152 -0.936  0.34922
## Season 6:   S = 0  -16 1096.7 -0.076 -0.453  0.65058
## Season 7:   S = 0   -8 1096.7 -0.038 -0.211  0.83259
## Season 8:   S = 0  -26 1096.7 -0.124 -0.755  0.45030
## Season 9:   S = 0   -2 1096.7 -0.010 -0.030  0.97591
## Season 10:   S = 0   6 1096.7  0.029  0.151  0.87999
## Season 11:   S = 0 -34 1096.7 -0.162 -0.996  0.31901
## Season 12:   S = 0 -28 1096.7 -0.133 -0.815  0.41489
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```r
# Dhaka PM2.5 not significant (p = 0.1068)

# Dhaka O3
Dhaka.Ozone.monthly.trend <- trend::smk.test(Dhaka.Ozone.monthly.ts)
Dhaka.Ozone.monthly.trend
```

```
##
##  Seasonal Mann-Kendall trend test (Hirsch-Slack test)
##
## data:  Dhaka.Ozone.monthly.ts
## z = -4.8206, p-value = 1.432e-06
## alternative hypothesis: true S is not equal to 0
## sample estimates:
##     S  varS
##  -554 13160
```

```r
summary(Dhaka.Ozone.monthly.trend)
```

```
##
##  Seasonal Mann-Kendall trend test (Hirsch-Slack test)
##
## data: Dhaka.Ozone.monthly.ts
## alternative hypothesis: two.sided
##
## Statistics for individual seasons
##
## H0
##                      S   varS    tau      z Pr(>|z|)
## Season 1:   S = 0  -30 1096.7 -0.143 -0.876 0.381187
## Season 2:   S = 0  -38 1096.7 -0.181 -1.117 0.263872
## Season 3:   S = 0    6 1096.7  0.029  0.151 0.879988
## Season 4:   S = 0  -56 1096.7 -0.267 -1.661 0.096747 .
## Season 5:   S = 0  -54 1096.7 -0.257 -1.600 0.109502
## Season 6:   S = 0  -40 1096.7 -0.190 -1.178 0.238924
## Season 7:   S = 0  -70 1096.7 -0.333 -2.084 0.037198 *
## Season 8:   S = 0  -76 1096.7 -0.362 -2.265 0.023527 *
## Season 9:   S = 0  -62 1096.7 -0.295 -1.842 0.065473 .
## Season 10:   S = 0 -12 1096.7 -0.057 -0.332 0.739764
## Season 11:   S = 0 -66 1096.7 -0.314 -1.963 0.049669 *
## Season 12:   S = 0 -56 1096.7 -0.267 -1.661 0.096747 .
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```r
# Dhaka ozone (O3) is significant (1.432e-06)

# Seasonal Mann-Kendall Delhi

# Delhi AQI
Delhi.AQI.monthly.trend <- trend::smk.test(Delhi.AQI.monthly.ts)
Delhi.AQI.monthly.trend
```

```
##
```

```
##  Seasonal Mann-Kendall trend test (Hirsch-Slack test)
##
## data:  Delhi.AQI.monthly.ts
## z = -0.98905, p-value = 0.3226
## alternative hypothesis: true S is not equal to 0
## sample estimates:
##        S     varS
##  -114.00 13053.33
```

```r
summary(Delhi.AQI.monthly.trend)
```

```
##
##  Seasonal Mann-Kendall trend test (Hirsch-Slack test)
##
## data: Delhi.AQI.monthly.ts
## alternative hypothesis: two.sided
##
## Statistics for individual seasons
##
## H0
##                      S   varS    tau      z Pr(>|z|)
## Season 1:   S = 0   -8 1092.0 -0.038 -0.212  0.83224
## Season 2:   S = 0  -41 1088.3 -0.199 -1.212  0.22532
## Season 3:   S = 0    1 1085.7  0.005  0.000  1.00000
## Season 4:   S = 0   -6 1092.0 -0.029 -0.151  0.87973
## Season 5:   S = 0  -31 1088.3 -0.150 -0.909  0.36316
## Season 6:   S = 0    9 1083.7  0.044  0.243  0.80799
## Season 7:   S = 0   16 1090.0  0.077  0.454  0.64959
## Season 8:   S = 0  -27 1088.3 -0.131 -0.788  0.43063
## Season 9:   S = 0    3 1083.7  0.015  0.061  0.95155
## Season 10:   S = 0  -4 1092.0 -0.019 -0.091  0.92766
## Season 11:   S = 0 -36 1084.7 -0.176 -1.063  0.28791
## Season 12:   S = 0  10 1084.7  0.049  0.273  0.78464
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```r
# Delhi AQI not significant (0.3226)

# Delhi PM2.5
Delhi.PM2.5.monthly.trend <- trend::smk.test(Delhi.PM2.5.monthly.ts)
Delhi.PM2.5.monthly.trend
```

```
##
##  Seasonal Mann-Kendall trend test (Hirsch-Slack test)
##
## data:  Delhi.PM2.5.monthly.ts
## z = -1.7156, p-value = 0.08624
## alternative hypothesis: true S is not equal to 0
## sample estimates:
##        S     varS
##  -197.00 13052.33
```

```r
summary(Delhi.PM2.5.monthly.trend)
```

```
##
##  Seasonal Mann-Kendall trend test (Hirsch-Slack test)
##
## data: Delhi.PM2.5.monthly.ts
## alternative hypothesis: two.sided
##
## Statistics for individual seasons
##
## H0
##                      S   varS    tau      z Pr(>|z|)
## Season 1:   S = 0  -24 1092.0 -0.115 -0.696  0.48642
## Season 2:   S = 0  -41 1088.3 -0.199 -1.212  0.22532
## Season 3:   S = 0  -13 1085.7 -0.063 -0.364  0.71571
## Season 4:   S = 0  -14 1092.0 -0.067 -0.393  0.69403
## Season 5:   S = 0  -19 1088.3 -0.092 -0.546  0.58533
## Season 6:   S = 0    3 1083.7  0.015  0.061  0.95155
## Season 7:   S = 0   -4 1090.0 -0.019 -0.091  0.92760
## Season 8:   S = 0  -13 1088.3 -0.063 -0.364  0.71605
## Season 9:   S = 0   -1 1083.7 -0.005  0.000  1.00000
## Season 10:   S = 0 -24 1092.0 -0.115 -0.696  0.48642
## Season 11:   S = 0 -37 1083.7 -0.181 -1.094  0.27413
## Season 12:   S = 0 -10 1084.7 -0.049 -0.273  0.78464
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```r
# Delhi AQI not significant (0.08624)

# Delhi PM10
Delhi.PM10.monthly.trend <- trend::smk.test(Delhi.PM10.monthly.ts)
Delhi.PM10.monthly.trend
```

```
##
##  Seasonal Mann-Kendall trend test (Hirsch-Slack test)
##
## data:  Delhi.PM10.monthly.ts
## z = -0.79649, p-value = 0.4257
## alternative hypothesis: true S is not equal to 0
## sample estimates:
##        S     varS
##   -92.00 13053.33
```

```r
summary(Delhi.PM10.monthly.trend)
```

```
##
##  Seasonal Mann-Kendall trend test (Hirsch-Slack test)
##
## data: Delhi.PM10.monthly.ts
## alternative hypothesis: two.sided
##
```

```
## Statistics for individual seasons
##
## H0
##                       S   varS    tau      z Pr(>|z|)
## Season 1:   S = 0   -2 1092.0 -0.010 -0.030  0.97586
## Season 2:   S = 0  -31 1088.3 -0.150 -0.909  0.36316
## Season 3:   S = 0   -5 1085.7 -0.024 -0.121  0.90338
## Season 4:   S = 0    2 1092.0  0.010  0.030  0.97586
## Season 5:   S = 0  -25 1088.3 -0.121 -0.727  0.46692
## Season 6:   S = 0    7 1083.7  0.034  0.182  0.85537
## Season 7:   S = 0   16 1090.0  0.077  0.454  0.64959
## Season 8:   S = 0  -31 1088.3 -0.150 -0.909  0.36316
## Season 9:   S = 0   -5 1083.7 -0.024 -0.122  0.90329
## Season 10:   S = 0   2 1092.0  0.010  0.030  0.97586
## Season 11:   S = 0 -30 1084.7 -0.146 -0.881  0.37857
## Season 12:   S = 0  10 1084.7  0.049  0.273  0.78464
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```r
# Delhi PM10 not significant (0.4257)

# Delhi O3
Delhi.Ozone.monthly.trend <- trend::smk.test(Delhi.Ozone.monthly.ts)
Delhi.Ozone.monthly.trend
```

```
##
##  Seasonal Mann-Kendall trend test (Hirsch-Slack test)
##
## data:  Delhi.Ozone.monthly.ts
## z = 1.0157, p-value = 0.3098
## alternative hypothesis: true S is not equal to 0
## sample estimates:
##        S     varS
##   117.00 13044.33
```

```r
summary(Delhi.Ozone.monthly.trend)
```

```
##
##  Seasonal Mann-Kendall trend test (Hirsch-Slack test)
##
## data: Delhi.Ozone.monthly.ts
## alternative hypothesis: two.sided
##
## Statistics for individual seasons
##
## H0
##                       S   varS    tau      z Pr(>|z|)
## Season 1:   S = 0   25 1091.0  0.120  0.727  0.46747
## Season 2:   S = 0   11 1088.3  0.053  0.303  0.76180
## Season 3:   S = 0   14 1084.7  0.068  0.395  0.69305
## Season 4:   S = 0   19 1091.0  0.092  0.545  0.58578
## Season 5:   S = 0    9 1088.3  0.044  0.242  0.80839
## Season 6:   S = 0   14 1082.7  0.069  0.395  0.69278
```

```
## Season 7:   S = 0   21 1089.0  0.102  0.606  0.54447
## Season 8:   S = 0    8 1087.3  0.039  0.212  0.83189
## Season 9:   S = 0    9 1083.7  0.044  0.243  0.80799
## Season 10:   S = 0   5 1091.0  0.024  0.121  0.90361
## Season 11:   S = 0  -7 1083.7 -0.034 -0.182  0.85537
## Season 12:   S = 0 -11 1083.7 -0.054 -0.304  0.76130
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```r
# Delhi ozone (O3) not significant (0.3098)


# Non-Seasonal Mann-Kendall Dhaka

# Dhaka AQI
Dhaka.monthly.AQI.nonseasonal <- Dhaka.AQI.monthly.ts -
  Dhaka.AQI.monthly.decomposed$time.series[,1]

Dhaka.nonseasonal.AQI.trend <- Kendall::MannKendall(
  Dhaka.monthly.AQI.nonseasonal)
Dhaka.nonseasonal.AQI.trend
```

```
## tau = -0.0537, 2-sided pvalue =0.20448
```

```r
summary(Dhaka.nonseasonal.AQI.trend)
```

```
## Score =  -1698 , Var(Score) = 1788626
## denominator =  31626
## tau = -0.0537, 2-sided pvalue =0.20448
```

```r
# Dhaka PM2.5
Dhaka.monthly.PM2.5.nonseasonal <- Dhaka.PM2.5.monthly.ts -
  Dhaka.PM2.5.monthly.decomposed$time.series[,1]

Dhaka.nonseasonal.PM2.5.trend <- Kendall::MannKendall(
  Dhaka.monthly.PM2.5.nonseasonal)
Dhaka.nonseasonal.PM2.5.trend
```

```
## tau = -0.0502, 2-sided pvalue =0.23537
```

```r
summary(Dhaka.nonseasonal.PM2.5.trend)
```

```
## Score =  -1588 , Var(Score) = 1788626
## denominator =  31626
## tau = -0.0502, 2-sided pvalue =0.23537
```

```r
# Dhaka PM10
Dhaka.monthly.PM10.nonseasonal <- Dhaka.PM10.monthly.ts -
  Dhaka.PM10.monthly.decomposed$time.series[,1]

Dhaka.nonseasonal.PM10.trend <- Kendall::MannKendall(
  Dhaka.monthly.PM10.nonseasonal)
Dhaka.nonseasonal.PM10.trend
```

```
## tau = -0.0502, 2-sided pvalue =0.23537
```

```r
summary(Dhaka.nonseasonal.PM10.trend)
```

```
## Score =  -1588 , Var(Score) = 1788626
## denominator =  31626
## tau = -0.0502, 2-sided pvalue =0.23537
```

```r
# Dhaka O3
Dhaka.monthly.Ozone.nonseasonal <- Dhaka.Ozone.monthly.ts -
  Dhaka.Ozone.monthly.decomposed$time.series[,1]

Dhaka.nonseasonal.Ozone.trend <- Kendall::MannKendall(
  Dhaka.monthly.Ozone.nonseasonal)
Dhaka.nonseasonal.Ozone.trend
```

```
## tau = -0.201, 2-sided pvalue =2.0616e-06
```

```r
summary(Dhaka.nonseasonal.Ozone.trend)
```

```
## Score =  -6350 , Var(Score) = 1788626
## denominator =  31626
## tau = -0.201, 2-sided pvalue =2.0616e-06
```

```r
# Non-Seasonal Mann-Kendall Delhi

# Delhi AQI
Delhi.monthly.AQI.nonseasonal <- Delhi.AQI.monthly.ts -
  Delhi.AQI.monthly.decomposed$time.series[,1]

Delhi.nonseasonal.AQI.trend <- Kendall::MannKendall(
  Delhi.monthly.AQI.nonseasonal)
Delhi.nonseasonal.AQI.trend
```

```
## tau = -0.0681, 2-sided pvalue =0.10775
```

```r
summary(Delhi.nonseasonal.AQI.trend)
```

```
## Score =  -2152 , Var(Score) = 1788519
## denominator =  31580.97
## tau = -0.0681, 2-sided pvalue =0.10775
```

```r
# Delhi PM2.5
Delhi.monthly.PM2.5.nonseasonal <- Delhi.PM2.5.monthly.ts -
  Delhi.PM2.5.monthly.decomposed$time.series[,1]

Delhi.nonseasonal.PM2.5.trend <- Kendall::MannKendall(
  Delhi.monthly.PM2.5.nonseasonal)
Delhi.nonseasonal.PM2.5.trend
```

```
## tau = -0.0903, 2-sided pvalue =0.03296
```

```r
summary(Delhi.nonseasonal.PM2.5.trend)
```

```
## Score =  -2853 , Var(Score) = 1788518
## denominator =  31580.47
## tau = -0.0903, 2-sided pvalue =0.03296
```

```r
# Delhi PM10
Delhi.monthly.PM10.nonseasonal <- Delhi.PM10.monthly.ts -
  Delhi.PM10.monthly.decomposed$time.series[,1]

Delhi.nonseasonal.PM10.trend <- Kendall::MannKendall(
  Delhi.monthly.PM10.nonseasonal)
Delhi.nonseasonal.PM10.trend
```

```
## tau = -0.0624, 2-sided pvalue =0.14094
```

```r
summary(Delhi.nonseasonal.PM10.trend)
```

```
## Score =  -1970 , Var(Score) = 1788519
## denominator =  31580.97
## tau = -0.0624, 2-sided pvalue =0.14094
```

```r
# Delhi O3
Delhi.monthly.Ozone.nonseasonal <- Delhi.Ozone.monthly.ts -
  Delhi.Ozone.monthly.decomposed$time.series[,1]

Delhi.nonseasonal.Ozone.trend <- Kendall::MannKendall(
  Delhi.monthly.Ozone.nonseasonal)
Delhi.nonseasonal.Ozone.trend
```

```
## tau = 0.0326, 2-sided pvalue =0.44208
```

```r
summary(Delhi.nonseasonal.Ozone.trend)
```

```
## Score =  1029 , Var(Score) = 1788510
## denominator =  31576.46
## tau = 0.0326, 2-sided pvalue =0.44208
```

Table 1: Summary of Mann-Kendall Results

Location	Variable	Seasonal p-value	Non-Seasonal p-value
Dhaka	AQI	0.0768	0.2045
Dhaka	PM 2.5	0.0768	0.2354
Dhaka	PM 10	0.1068	0.2354
Dhaka	O3	**1.43e-06**	**2.06e-06**
Delhi	AQI	0.3226	0.1078
Delhi	PM 2.5	0.0862	**0.0330**
Delhi	PM 10	0.4257	0.1409

Location	Variable	Seasonal p-value	Non-Seasonal p-value
Delhi	O3	0.3098	0.4421

```r
# AQI
t.test(AQI ~ Location, data = combined_dataset)
```

```
##
##  Welch Two Sample t-test
##
## data:  AQI by Location
## t = 19.658, df = 344.72, p-value < 2.2e-16
## alternative hypothesis: true difference in means between group Delhi and group Dhaka is not equal to
## 95 percent confidence interval:
##   94.82248 115.90689
## sample estimates:
## mean in group Delhi mean in group Dhaka
##            275.7138            170.3491
```

```r
# PM2.5
t.test(PM2.5 ~ Location, data = combined_dataset)
```

```
##
##  Welch Two Sample t-test
##
## data:  PM2.5 by Location
## t = 35.011, df = 490.03, p-value < 2.2e-16
## alternative hypothesis: true difference in means between group Delhi and group Dhaka is not equal to
## 95 percent confidence interval:
##  132.4543 148.2050
## sample estimates:
## mean in group Delhi mean in group Dhaka
##            240.5413            100.2117
```

```r
# PM10
t.test(PM10 ~ Location, data = combined_dataset)
```

```
##
##  Welch Two Sample t-test
##
## data:  PM10 by Location
## t = 10.773, df = 490.34, p-value < 2.2e-16
## alternative hypothesis: true difference in means between group Delhi and group Dhaka is not equal to
## 95 percent confidence interval:
##  59.67191 86.29319
## sample estimates:
## mean in group Delhi mean in group Dhaka
##            242.1669            169.1844
```

```r
# O3
t.test(O3 ~ Location, data = combined_dataset)
```

```
##
##  Welch Two Sample t-test
##
## data:  O3 by Location
## t = -15.654, df = 450.94, p-value < 2.2e-16
## alternative hypothesis: true difference in means between group Delhi and group Dhaka is not equal to
## 95 percent confidence interval:
##  -21.48934 -16.69565
## sample estimates:
## mean in group Delhi mean in group Dhaka
##            31.78149            50.87398
```

Table 2: Summary of two sample t-test Results

Variable	p-value	Dhaka mean	Delhi mean
AQI	**2.2e-16**	170.35	275.71
PM2.5	**2.2e-16**	100.21	240.54
PM10	**2.2e-16**	169.18	242.17
O3	**2.2e-16**	50.87	31.78

```r
#Linear Regression

# AQI
AQI.model <- lm(AQI ~ first.month * Location, data = combined_dataset)
summary(AQI.model)
```

```
##
## Call:
## lm(formula = AQI ~ first.month * Location, data = combined_dataset)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -152.894  -19.418   -1.296   16.357   93.331
##
## Coefficients:
##                            Estimate Std. Error t value Pr(>|t|)
## (Intercept)               7.595e+02  1.582e+01   48.00   <2e-16 ***
## first.month              -3.048e-02  9.873e-04  -30.87   <2e-16 ***
## LocationDhaka            -7.062e+02  2.238e+01  -31.56   <2e-16 ***
## first.month:LocationDhaka  3.785e-02  1.396e-03   27.11   <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 34.7 on 500 degrees of freedom
## Multiple R-squared:  0.8128, Adjusted R-squared:  0.8117
## F-statistic: 723.5 on 3 and 500 DF,  p-value: < 2.2e-16
```

```r
# PM2.5
PM2.5.model <- lm(PM2.5 ~ first.month * Location, data = combined_dataset)
summary(PM2.5.model)
```

```
##
## Call:
## lm(formula = PM2.5 ~ first.month * Location, data = combined_dataset)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -127.439  -13.193   -3.511    9.854   79.986
##
## Coefficients:
##                             Estimate Std. Error t value Pr(>|t|)
## (Intercept)                5.623e+02  1.238e+01   45.42   <2e-16 ***
## first.month               -2.027e-02  7.725e-04  -26.24   <2e-16 ***
## LocationDhaka             -6.310e+02  1.751e+01  -36.04   <2e-16 ***
## first.month:LocationDhaka  3.091e-02  1.092e-03   28.30   <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 27.15 on 500 degrees of freedom
## Multiple R-squared:  0.8946, Adjusted R-squared:  0.894
## F-statistic:  1415 on 3 and 500 DF,  p-value: < 2.2e-16
```

```r
# PM10
PM10.model <- lm(PM10 ~ first.month * Location, data = combined_dataset)
summary(PM10.model)
```

```
##
## Call:
## lm(formula = PM10 ~ first.month * Location, data = combined_dataset)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -214.902  -26.310   -5.708   23.492  128.747
##
## Coefficients:
##                             Estimate Std. Error t value Pr(>|t|)
## (Intercept)                7.438e+02  2.324e+01   32.00   <2e-16 ***
## first.month               -3.160e-02  1.450e-03  -21.79   <2e-16 ***
## LocationDhaka             -8.491e+02  3.287e+01  -25.83   <2e-16 ***
## first.month:LocationDhaka  4.890e-02  2.051e-03   23.84   <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 50.98 on 500 degrees of freedom
## Multiple R-squared:  0.6364, Adjusted R-squared:  0.6342
## F-statistic: 291.7 on 3 and 500 DF,  p-value: < 2.2e-16
```

```r
# O3
O3.model <- lm(O3 ~ first.month * Location, data = combined_dataset)
summary(O3.model)
```

```
##
## Call:
## lm(formula = O3 ~ first.month * Location, data = combined_dataset)
```

```
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -30.486  -9.189  -4.640  13.627  32.740
##
## Coefficients:
##                             Estimate Std. Error t value Pr(>|t|)
## (Intercept)               -2.075e+01  5.773e+00  -3.595 0.000356 ***
## first.month                3.310e-03  3.602e-04   9.189  < 2e-16 ***
## LocationDhaka              6.263e+01  8.164e+00   7.671 8.97e-14 ***
## first.month:LocationDhaka -2.743e-03  5.094e-04  -5.384 1.12e-07 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 12.66 on 500 degrees of freedom
## Multiple R-squared:  0.4275, Adjusted R-squared:  0.4241
## F-statistic: 124.5 on 3 and 500 DF,  p-value: < 2.2e-16
```

Table 3: Summary of Linear Regression Results

Variable	p-value	Min	Median	Max	R-squared
AQI	**2.2e-16**	-152.894	-1.296	93.331	0.812
PM2.5	**2.2e-16**	-127.439	-3.511	79.986	0.894
PM10	**2.2e-16**	-214.902	-5.708	128.747	0.634
O3	**2.2e-16**	-30.487	-4.640	32.740	0.424